

Applied Statistics (MATH10096)

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Modelling rates: including an offset in the model

- ↪ Often, the expected value of a response count y_i is proportional to an index, say o_i .
- ↪ For instance, o_i might represent an amount of time or a population size, such as when modelling COVID-19 deaths across different countries.
- ↪ Alternatively, it could be a spatial area, such as when modelling the counts of a particular animal or plant species.
- ↪ In such cases, the sample rate is y_i/o_i , with expected value $\mathbb{E}(y_i/o_i) = \mathbb{E}(y_i)/o_i = \mu_i/o_i$.

Modelling rates: including an offset in the model

- ↪ When predictor variables are included, a loglinear model for the expected rate takes the form

$$\log \left(\frac{\mu_i}{o_i} \right) = \mathbf{x}_i^T \boldsymbol{\beta}, \quad i = 1, \dots, n.$$

- ↪ Thus

$$\log \mu_i = \log o_i + \mathbf{x}_i^T \boldsymbol{\beta}.$$

- ↪ The adjustment term $\log o_i$ is called an **offset**.
- ↪ The fit corresponds to using $\log o_i$ as a predictor in the linear predictor for $\log \mu_i$ with its coefficient fixed at one.

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

- ↪ Data on the survival and death of 539 males diagnosed with lung cancer.
- ↪ Two factors are of interest: histology and stage of disease, each grouped into three categories.
- ↪ The period of follow-up was divided into a series of two-month intervals.
- ↪ For each combination of follow-up interval, histology, and stage of disease, the number of deaths and the total follow-up time (in months) are recorded.
- ↪ We treat the death counts for each cell (i.e., each combination of factors and follow-up interval) as independent Poisson variates.

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

```
url <- "https://users.stat.ufl.edu/~aa/glm/data/Cancer.dat"
Cancer <- read.table(url, header = TRUE)
head(Cancer)
```

```
##   time histology stage count risktime
## 1    1         1     1     9      157
## 2    1         2     1     5       77
## 3    1         3     1     1       21
## 4    2         1     1     2      139
## 5    2         2     1     2       68
## 6    2         3     1     1       17
```

```
tail(Cancer)
```

```
##   time histology stage count risktime
## 58    6         1     3     4       39
## 59    6         2     3     3       13
## 60    6         3     3     3       14
## 61    7         1     3     1       29
## 62    7         2     3     2        7
## 63    7         3     3     3       10
```

→ Time 1 corresponds to the follow-up interval of 0–2 months (after diagnosis), time 2 to 2–4 months, and so on, with time 7 corresponding to the period 12+ months after diagnosis.

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

- ↪ Let μ_{ijk} denote the expected number of deaths and o_{ijk} the total time at risk for histology i and stage of disease j , in follow-up time interval k , for $i \in \{1, 2, 3\}$, $j \in \{1, 2, 3\}$, and $k \in \{1, 2, \dots, 7\}$.
- ↪ The Poisson loglinear model for the death rate is

$$\log \mu_{ijk} = \log o_{ijk} + \beta_0 + \beta_i^H + \beta_j^S + \beta_k^T.$$

- ↪ This model treats each predictor as a factor variable, where the superscript notation shows the classification labels (**H**istology, **S**tage, **T**ime).

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

↪ An offset for a regression model is specified in R by using the `offset` function.

```
fit_lc <- glm(count ~ factor(histology) + factor(stage) +  
              factor(time) + offset(log(risktime)),  
              family = poisson, data = Cancer)
```

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

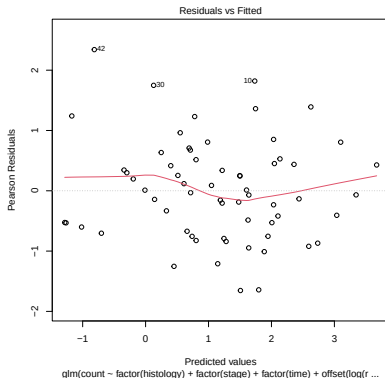
```
printCoefmat(coef(summary(fit_lc)), digits = 3)
```

```
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -3.0093     0.1665  -18.07   <2e-16 ***
## factor(histology)2    0.1624     0.1220    1.33   0.1829
## factor(histology)3    0.1075     0.1475    0.73   0.4658
## factor(stage)2       0.4700     0.1744    2.69   0.0071 **
## factor(stage)3       1.3243     0.1521    8.71   <2e-16 ***
## factor(time)2       -0.1275     0.1491   -0.85   0.3926
## factor(time)3       -0.0797     0.1635   -0.49   0.6258
## factor(time)4        0.1189     0.1711    0.70   0.4869
## factor(time)5       -0.6651     0.2606   -2.55   0.0107 *
## factor(time)6       -0.3502     0.2435   -1.44   0.1504
## factor(time)7       -0.1752     0.2499   -0.70   0.4832
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```


Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

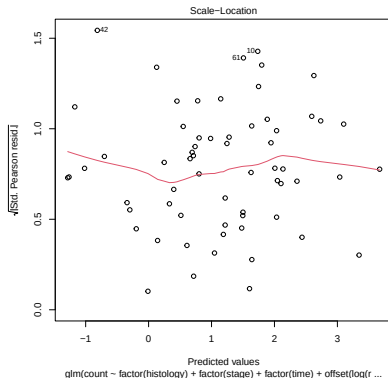
```
plot(fit_lc, which = 1)
```



Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

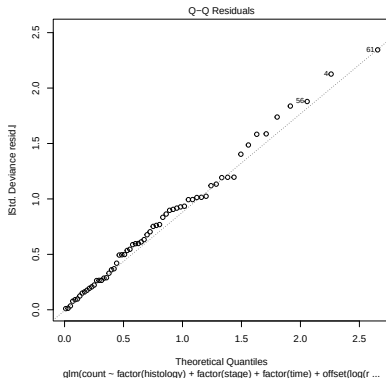
```
plot(fit_lc, which = 3)
```



Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

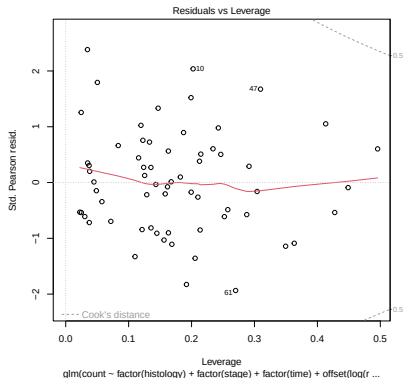
```
plot(fit_lc, which = 2)
```



Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

```
plot(fit_lc, which = 5)
```



Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

- ↪ The residual plots do not evidence any problem.
- ↪ A generalized likelihood ratio test further reveals that stage of disease is an important predictor, whereas histology is not.
- ↪ Additionally, the test does not support including an interaction between stage and time.

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

- ↪ How to interpret the coefficients in the case of a Poisson regression?
- ↪ Let us see how to interpret the effect of disease stage on the estimated death rate.
- ↪ The log death rate at the second stage of disease is given by

$$\log \left(\frac{\mu_{i2k}}{o_{i2k}} \right) = \beta_0 + \beta_i^H + \beta_2^S + \beta_k^T.$$

- ↪ The log death rate at the first stage of disease is

$$\log \left(\frac{\mu_{i1k}}{o_{i1k}} \right) = \beta_0 + \beta_i^H + \beta_k^T.$$

since $\beta_1^S = 0$ for identifiability.

Modelling rates: including an offset in the model

Lung cancer example (Agresti, 2015, Chapter 7)

→ The difference in log death rates, keeping histology and follow-up interval the same, is

$$\log\left(\frac{\mu_{i2k}}{o_{i2k}}\right) - \log\left(\frac{\mu_{i1k}}{o_{i1k}}\right) = \beta_2^S.$$

→ Simple algebra then gives

$$\frac{\mu_{i2k}}{o_{i2k}} = \frac{\mu_{i1k}}{o_{i1k}} e^{\beta_2^S}.$$

→ The estimated death rate at the second stage of disease is $e^{\hat{\beta}_2^S} = e^{0.47} = 1.6$ times that at the first stage, adjusting for follow-up time and histology (i.e., holding these predictors fixed for both disease stages being compared), with 95% confidence interval $(e^{0.47-0.17 \times 1.96}, e^{0.47+0.17 \times 1.96}) = (1.147, 2.233)$.

→ Similarly, the estimated death rate at the third stage of disease is $e^{\hat{\beta}_3^S} = e^{1.32} = 3.74$ times that at the first stage, adjusting for follow-up time and histology, with the 95% CI being $(e^{1.32-0.15 \times 1.96}, e^{1.32+0.15 \times 1.96}) = (2.790, 5.023)$.

Gamma regression: Insurance claims testing

- ↪ The dataset `motori` is derived from the `motorins` dataset in the `faraway` package in R.
- ↪ It contains insurance company data from Sweden, on `Payment` of claims in a year, based on `Make` of car (a factor variable with nine levels representing different car models), number of years of no claims, `Bonus`, annual number of `km` driven (a numeric variable with five discrete values), and number of `Insured` in policy-years.
- ↪ Note that `Insured` is not an integer variable to account for the fact that not all insured individuals are insured for the entire year.

Gamma regression: Insurance claims testing

↪ An initial model for the data is:

$$\mathbb{E}(\text{Payment}_i) = \text{Insured}_i \times \text{risk}_i, \quad \text{Payment} \sim \text{gamma},$$

so

$$\log\{\mathbb{E}(\text{Payment}_i)\} = \log(\text{Insured}_i) + \log(\text{risk}_i).$$

↪ $\log(\text{Insured}_i)$ is included as an offset because the total amount of claims, and hence the payment amount, for a group is generally expected to be proportional to the number of insured individuals.

Gamma regression: Insurance claims testing

↪ $\log(\text{risk}_i)$ can be modelled using a linear model structure to give:

$$\log\{\mathbb{E}(\text{Payment}_i)\} = \log(\text{Insured}_i) + \alpha + \beta_j + \gamma_{\text{km}_i} + \delta_j \text{km}_i + \zeta_{\text{Bonus}_i}.$$

- ↪ Here β_j is the main effect for `Make`, $j \in \{1, \dots, 9\}$, γ_{km_i} is the main effect of kms traveled, $\delta_j \text{km}_i$ is the `Make-km` interaction, and ζ_{Bonus_i} is the effect of `Bonus`.
- ↪ Consider testing the null hypothesis that the effect of `kms` travelled on payment rate does not vary by car model (encoded in the `Make` variable).
- ↪ That is, testing whether the null hypothesis that the interaction term between the `Make` variable and `km` is not required in the model.

Gamma regression: Insurance claims testing

- First, let us fit the corresponding null and full models — that is, the models without and with the interaction term, respectively.

```
require(faraway)
data(motorins)

#attention will be restricted to data from zone 1
motori <- motorins[motorins$Zone==1,]
motori$km <- as.numeric(motori$Kilometres)
#we are not using the variables perd and claims
head(motori)

##   Kilometres Zone Bonus Make Insured Claims Payment      perd km
## 1           1     1     1     1   455.13     108  392491 3634.176 1
## 2           1     1     1     2    69.17      19   46221 2432.684 1
## 3           1     1     1     3    72.88      13   15694 1207.231 1
## 4           1     1     1     4 1292.39     124  422201 3404.847 1
## 5           1     1     1     5   191.01      40  119373 2984.325 1
## 6           1     1     1     6   477.66      57  170913 2998.474 1

#full model
g1 <- glm(Payment ~ offset(log(Insured)) + km*Make + Bonus,
          family = Gamma(link = log), data = motori)

#null model
g0 <- glm(Payment ~ offset(log(Insured)) + km + Make + Bonus,
          family = Gamma(link = log), data = motori)
```

Gamma regression: Insurance claims testing

- ↪ Since ϕ is not known for the gamma distribution, we now perform an F-ratio test comparing the models.

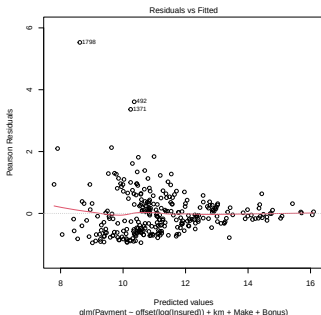
```
anova(g0, g1, test = "F")  
## Analysis of Deviance Table  
##  
## Model 1: Payment ~ offset(log(Insured)) + km + Make + Bonus  
## Model 2: Payment ~ offset(log(Insured)) + km * Make + Bonus  
##   Resid. Df Resid. Dev Df Deviance      F Pr(>F)  
## 1      284      155.06  
## 2      276      151.89  8      3.166 0.752 0.6455
```

- ↪ It appears that the dependence of the payment rate on km travelled does not vary with car make.

Gamma regression: Insurance claims testing

- The residual plots for both models, g_0 and g_1 , indicate that the mean–variance relationship implied by the Gamma distribution is not appropriate for these data, and hence all conclusions should be interpreted with caution.

```
plot(g0, which = 1)
```



Gamma regression: Insurance claims testing

```
plot(g0, which = 3)
```

